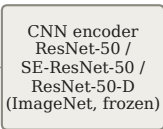




image
 256×256



$8 \times 8 \times 2048$
avg-pool to
 $7 \times 7 \times 2048$

linear
 $2048 \rightarrow 512$
(trained)

$\mathbf{a}_1, \dots, \mathbf{a}_{49}$
49 vectors in $\mathbb{R}^{512}$

attention LSTM decoder, step t (trained)

embedding
 $\mathbf{E}y_{t-1}$

gate
 $\boldsymbol{\beta}_t \odot \hat{\mathbf{z}}_t$

attention
 $\alpha_{t,k}, \hat{\mathbf{z}}_t$

LSTM
 $\mathbf{h}_t, \mathbf{c}_t$

linear +
softmax
 $p(y_t)$

y_t

■ frozen (pretrained)

■ trained

$\mathbf{h}_t$ used at step $t+1$